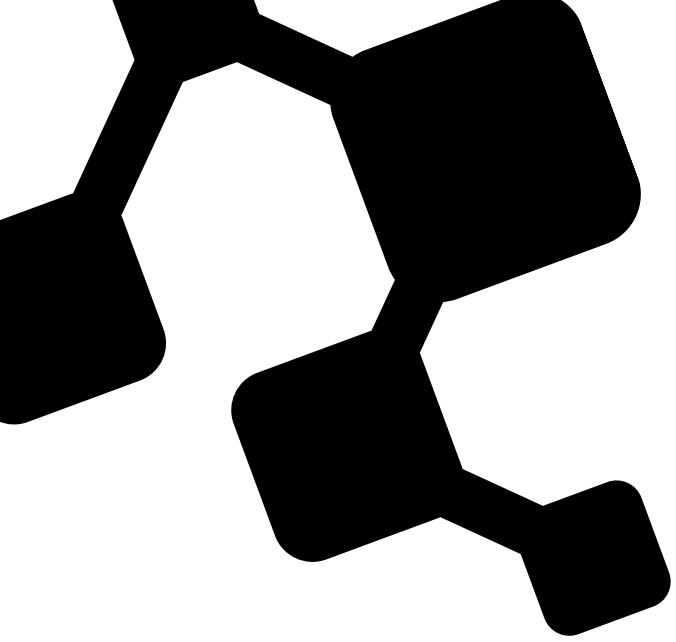
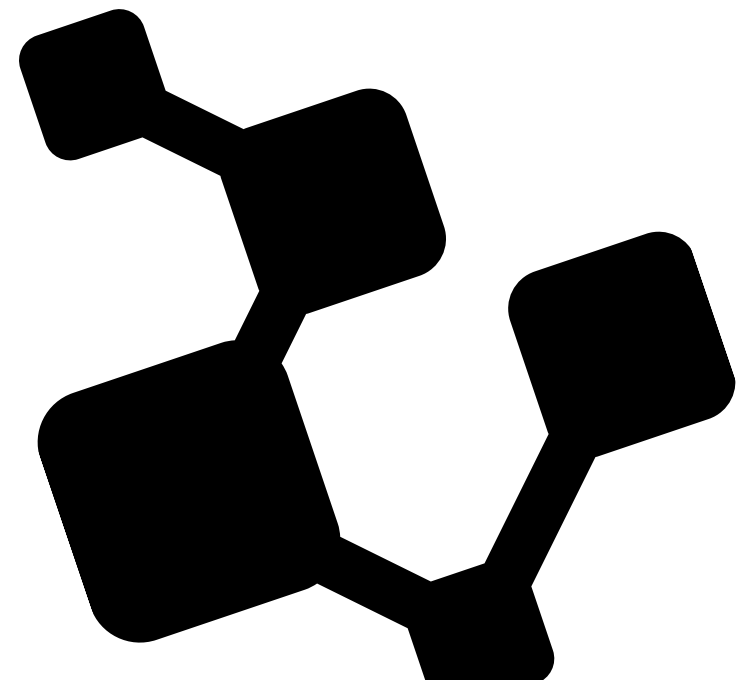
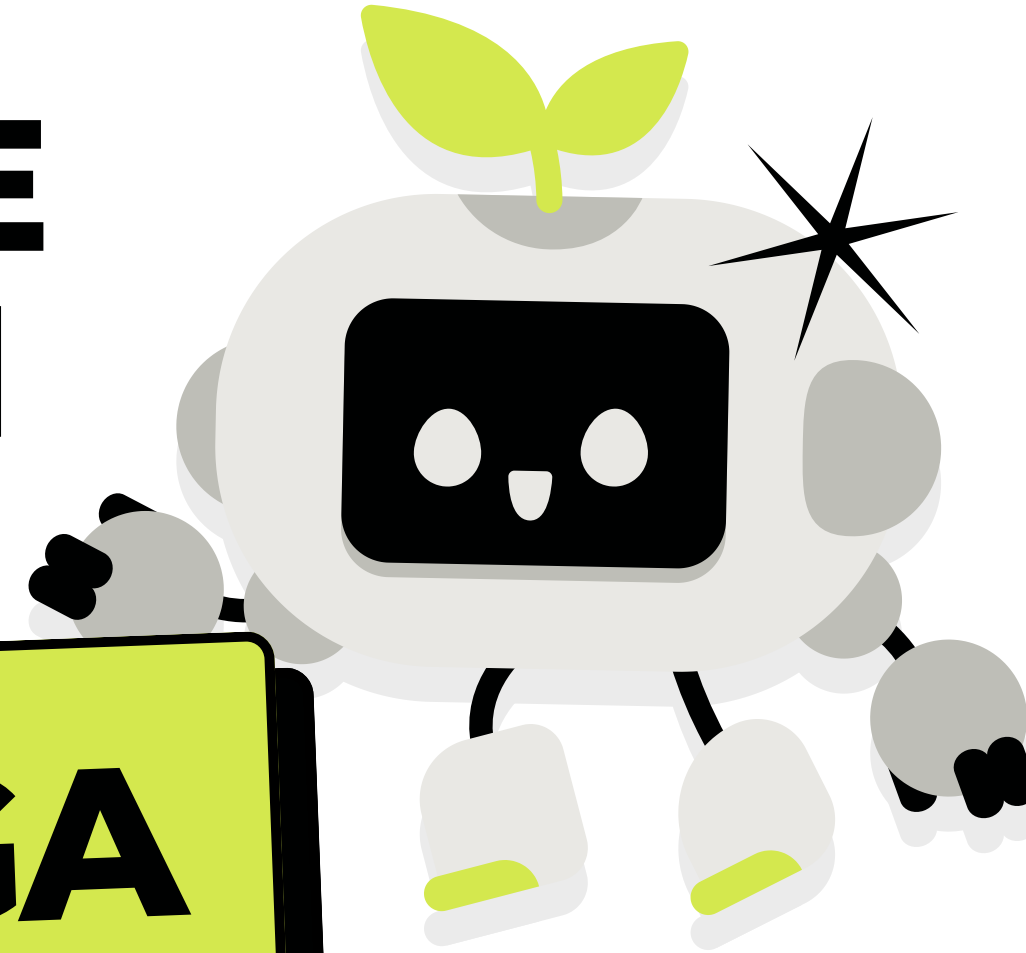
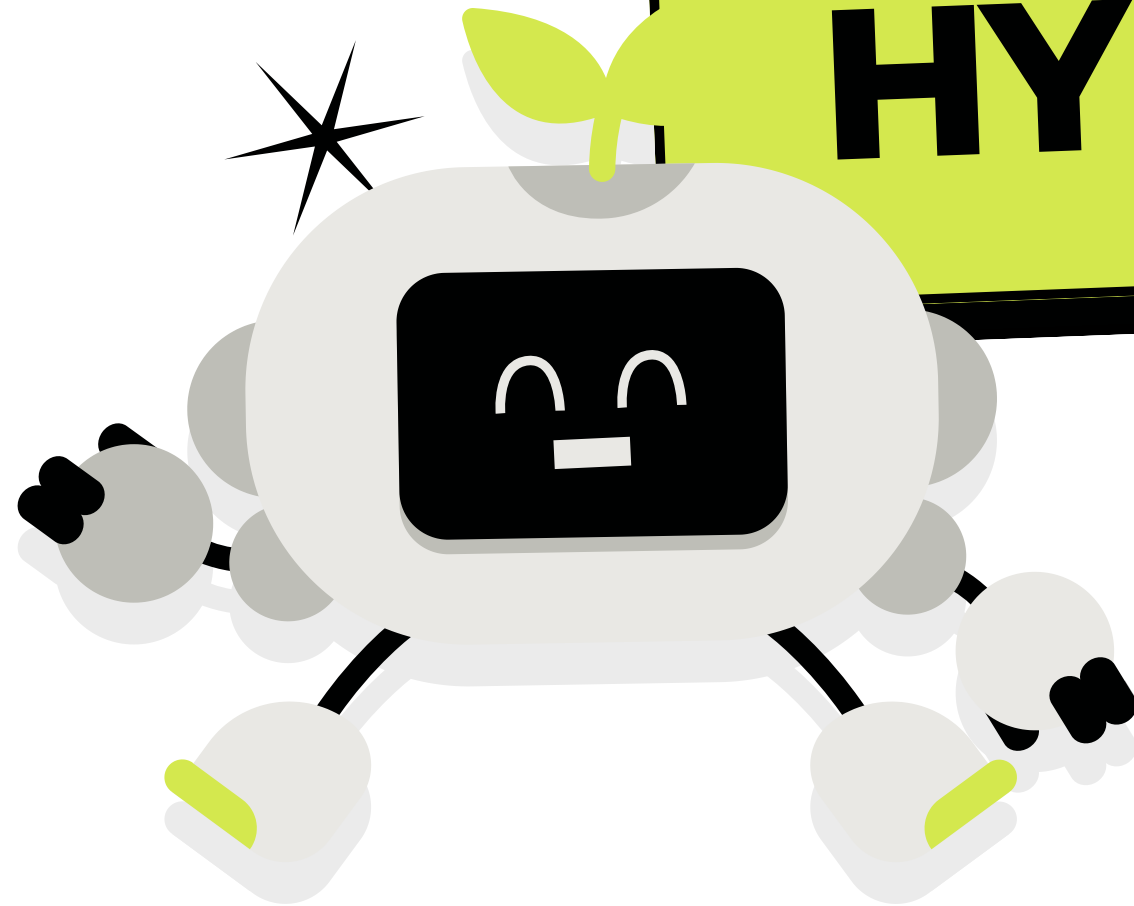


INTELLIGENT MOBILE ROBOT NAVIGATION USING

HYBRID FUZZY - GA

Presented by :

- | | |
|--------------------|---------|
| 1. ALYA SYAFINA | 2218084 |
| 2. NUR IRDINA | 2312378 |
| 3. WAN NUR ATHIRAH | 2320102 |



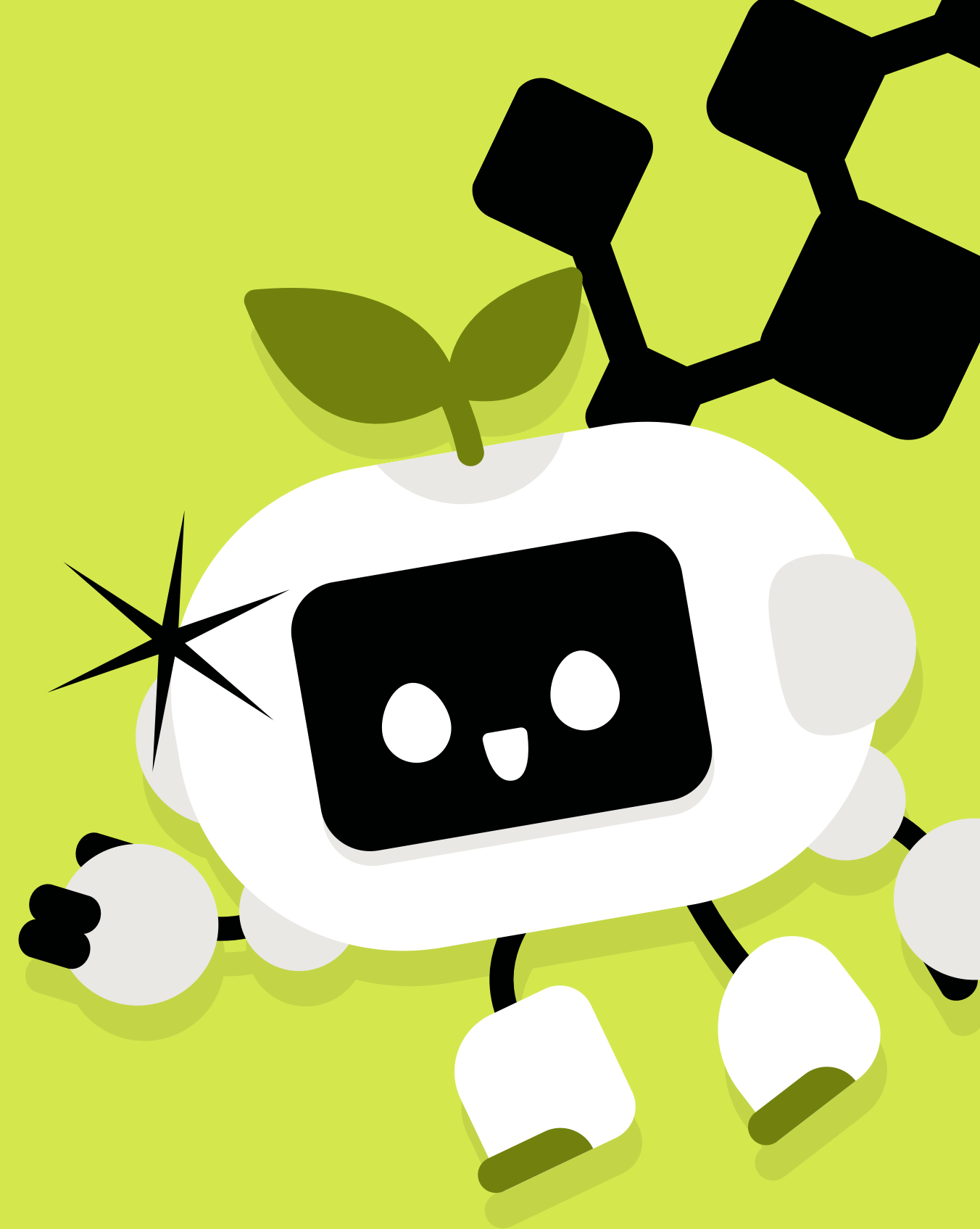
PROJECT SUMMARY

01 OBJECTIVE

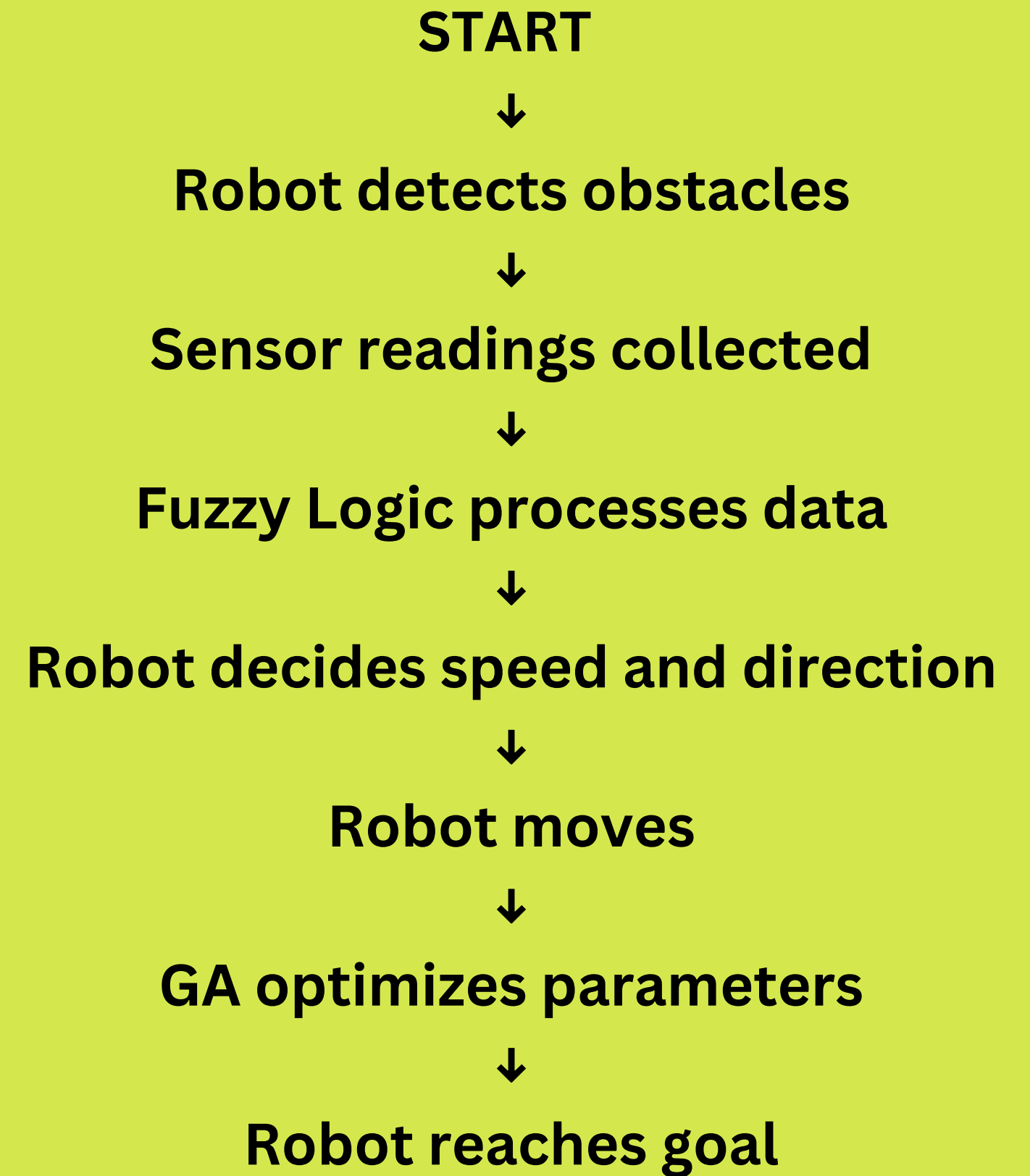
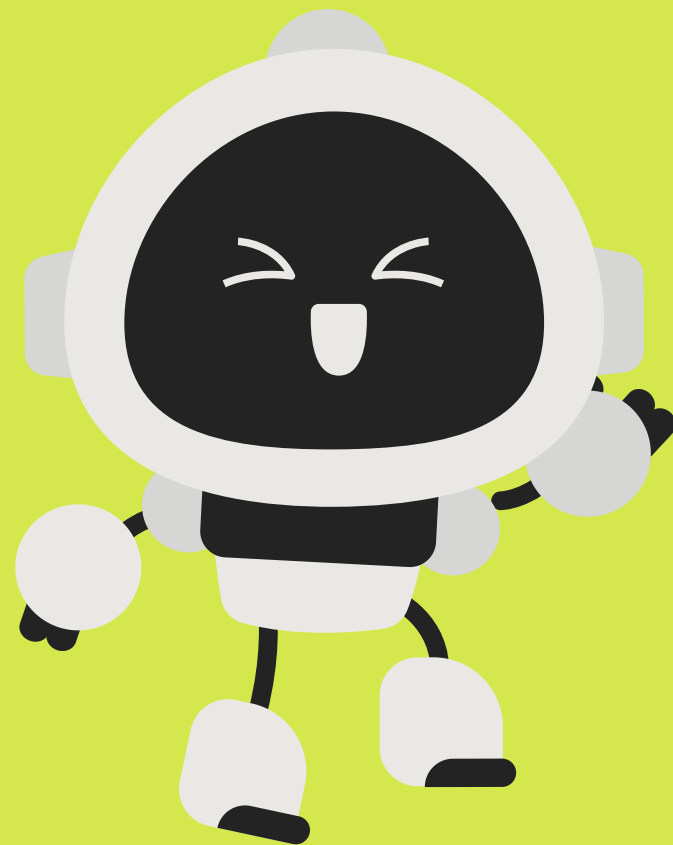
To build a robot navigation system that can move from a starting point to a target point. in an environment full of obstacles without collision.

02 WHY FUZZY - GA ?

Traditional logic struggles to handle constantly changing distances. Fuzzy logic is excellent for smoothly avoiding obstacles, while GA is used to automatically find the best settings for the robot's driving



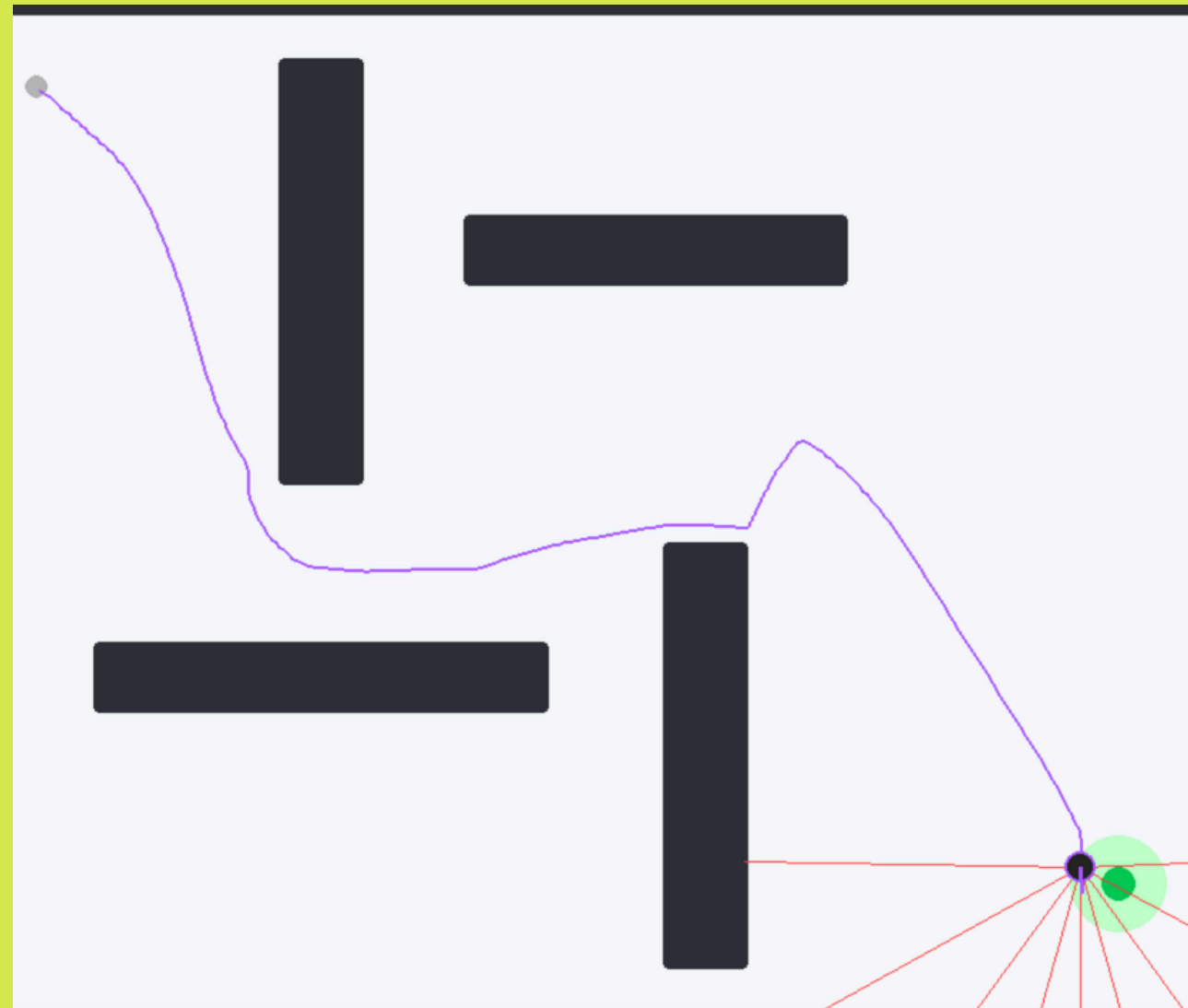
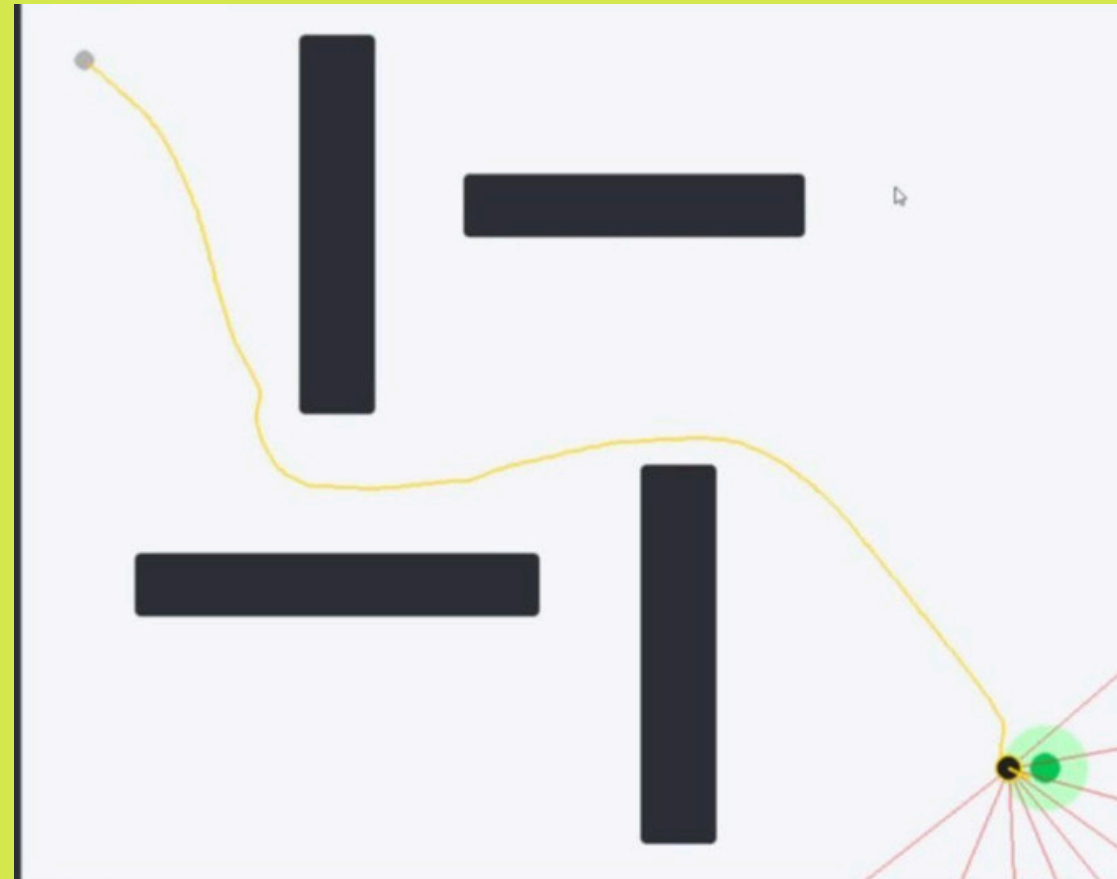
HOW THE ROBOT WORKS?



SIMULATION

Environment Setup

- Start Point (Grey Circle)
- Goal Point (Green Circle)
- Obstacles (Black Blocks)
- Virtual Sensors (Red Lines)
- Robot Path (Yellow/Purple Line)

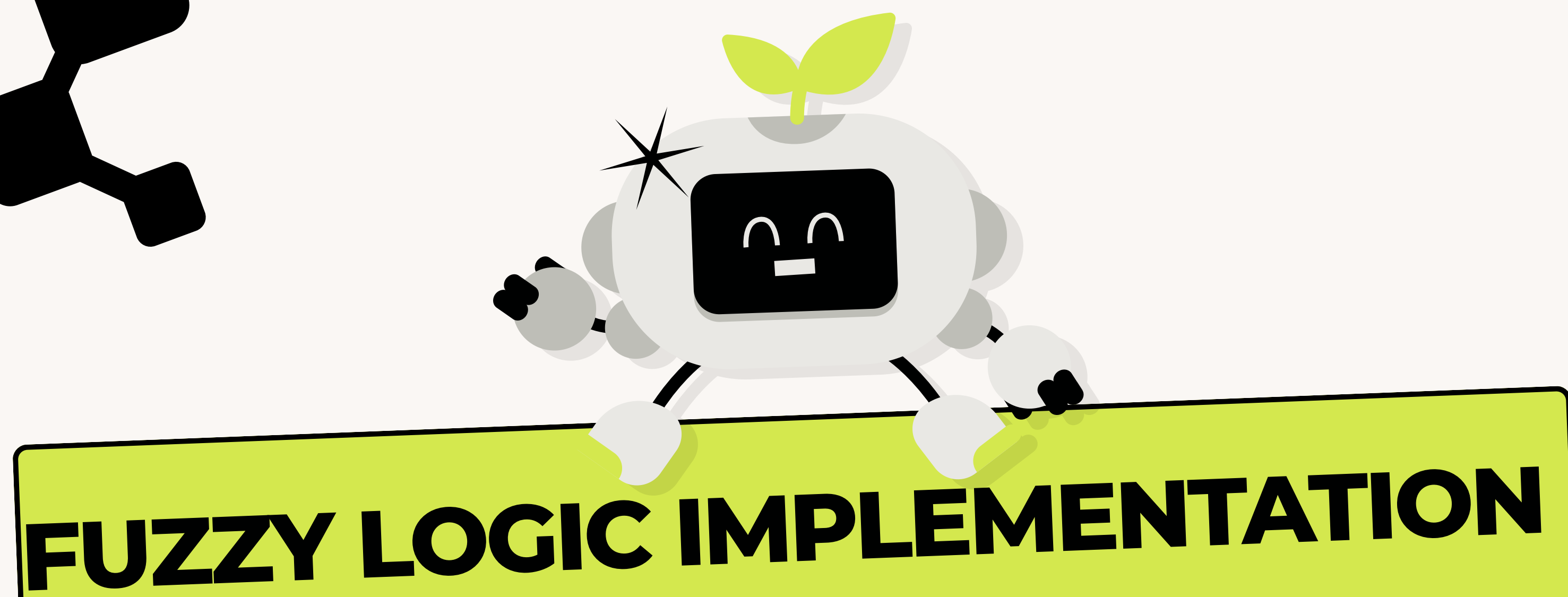


Controller Inputs

- Left Distance Sensor
- Front Distance Sensor
- Right Distance Sensor
- Goal Direction Angle

Robot Parameters

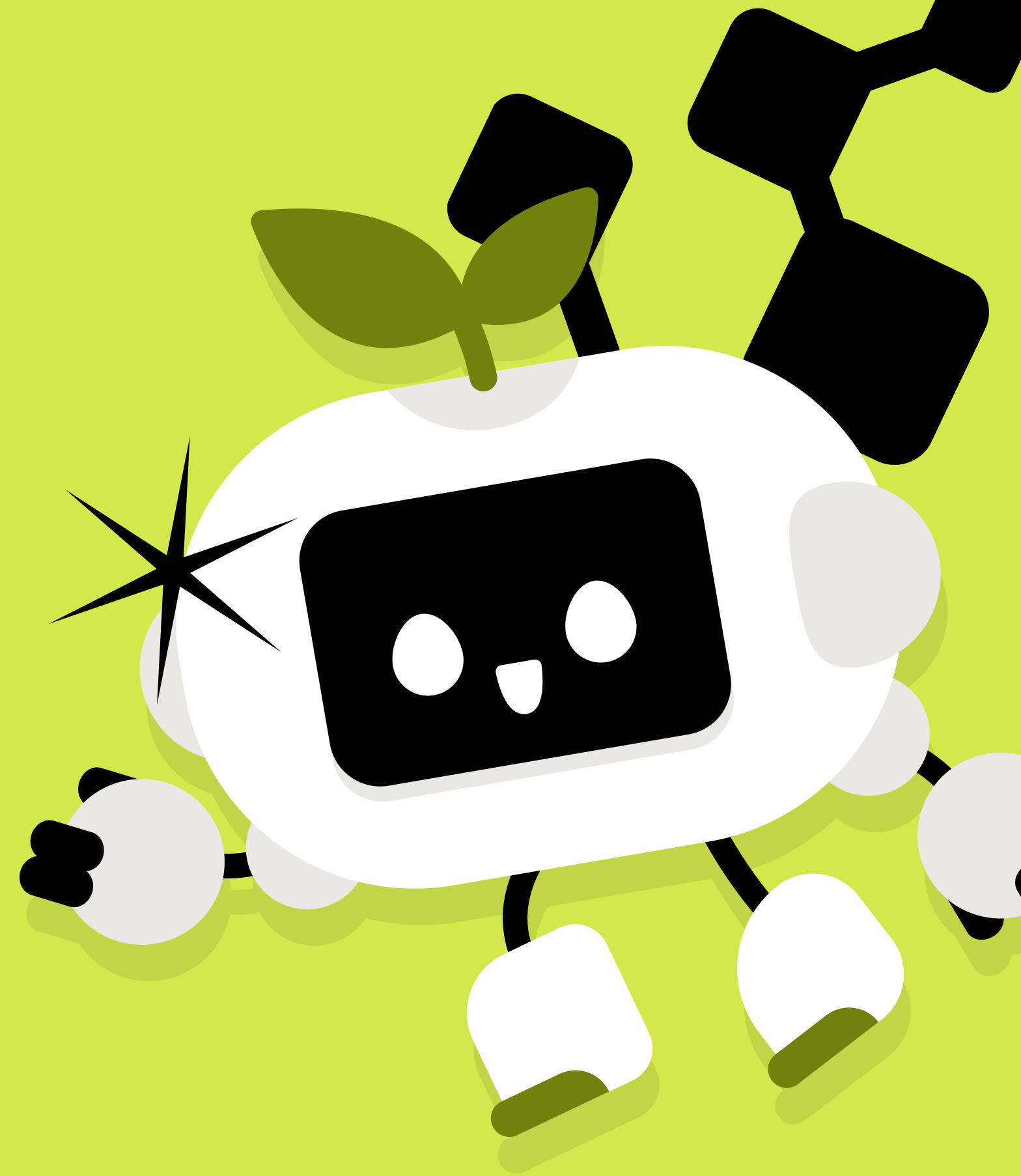
- Sensitivity
- Collisions
- Steps
- Distance
- Turn power
- Goal force
- Speed
- Escape



FUZZY LOGIC IMPLEMENTATION

FUZZY IN OUR SYSTEM

- **INPUTS** → left_value, front_value and right_value .
- **MEMBERSHIP FUNCTIONS** → Near/Far.
- **DEFUZIFICATION** → Turn and speed.

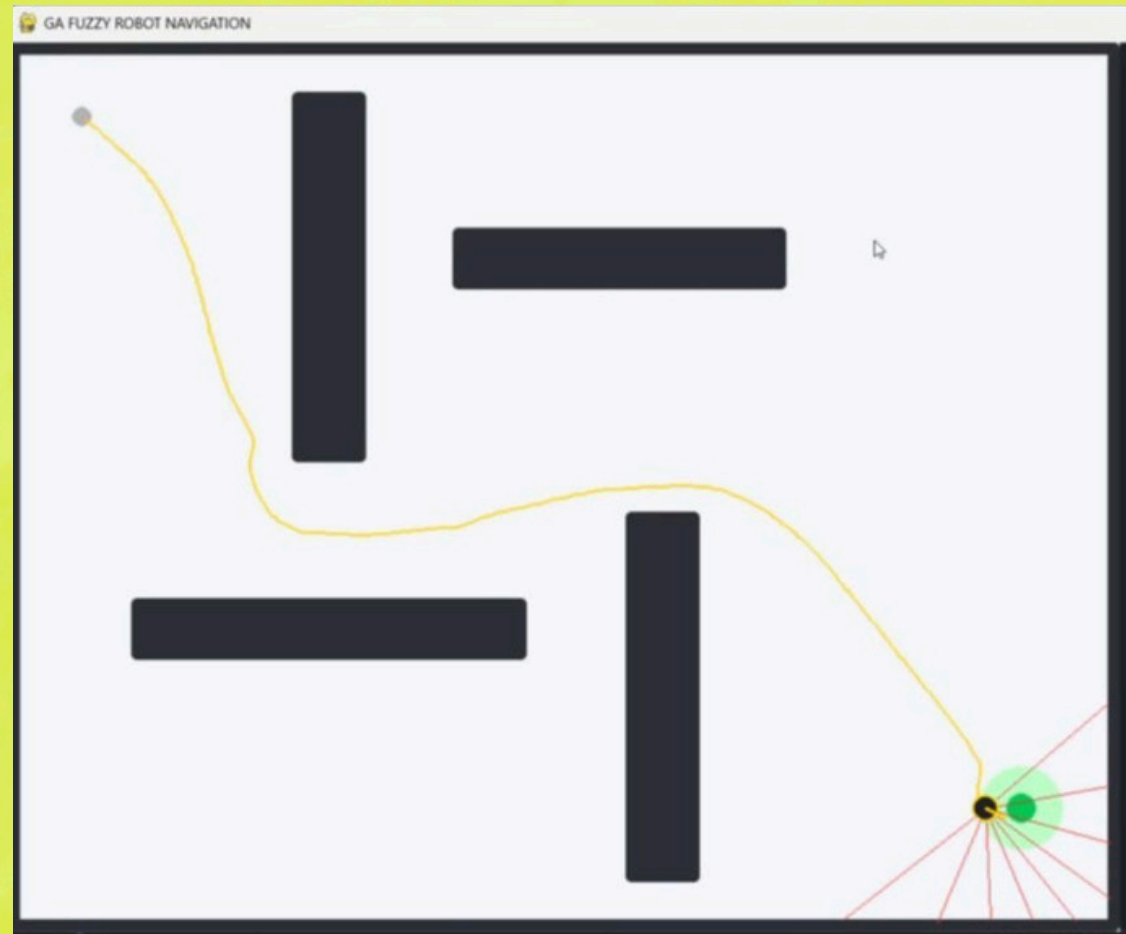


FUZZY RULES

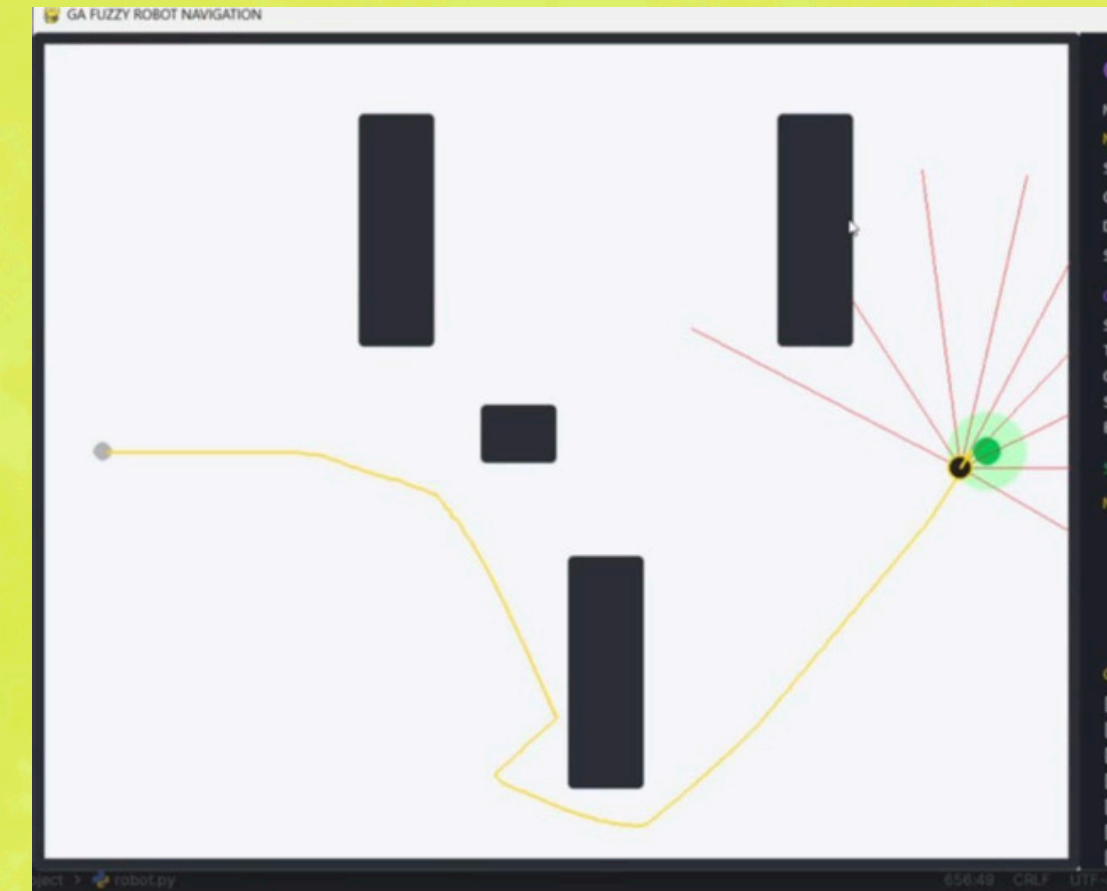
- 1) IF a wall is directly in front  THEN check for the biggest open space and steer toward it.
- 2) IF a wall gets close on the left  THEN turn the steering to the right.
- 3) IF a wall gets close on the right  THEN turn the steering to the left.
- 4) IF the robot is not facing the goal  THEN pull the steering toward the green target.
- 5) IF a wall is extremely close in front  THEN drop the speed to a slow crawl to prevent a crash.
- 6) IF the robot is surrounded by obstacles  THEN cut the speed in half to handle corners safely.

FUZZY LOGIC IN OUR SIMULATION

MAP 1

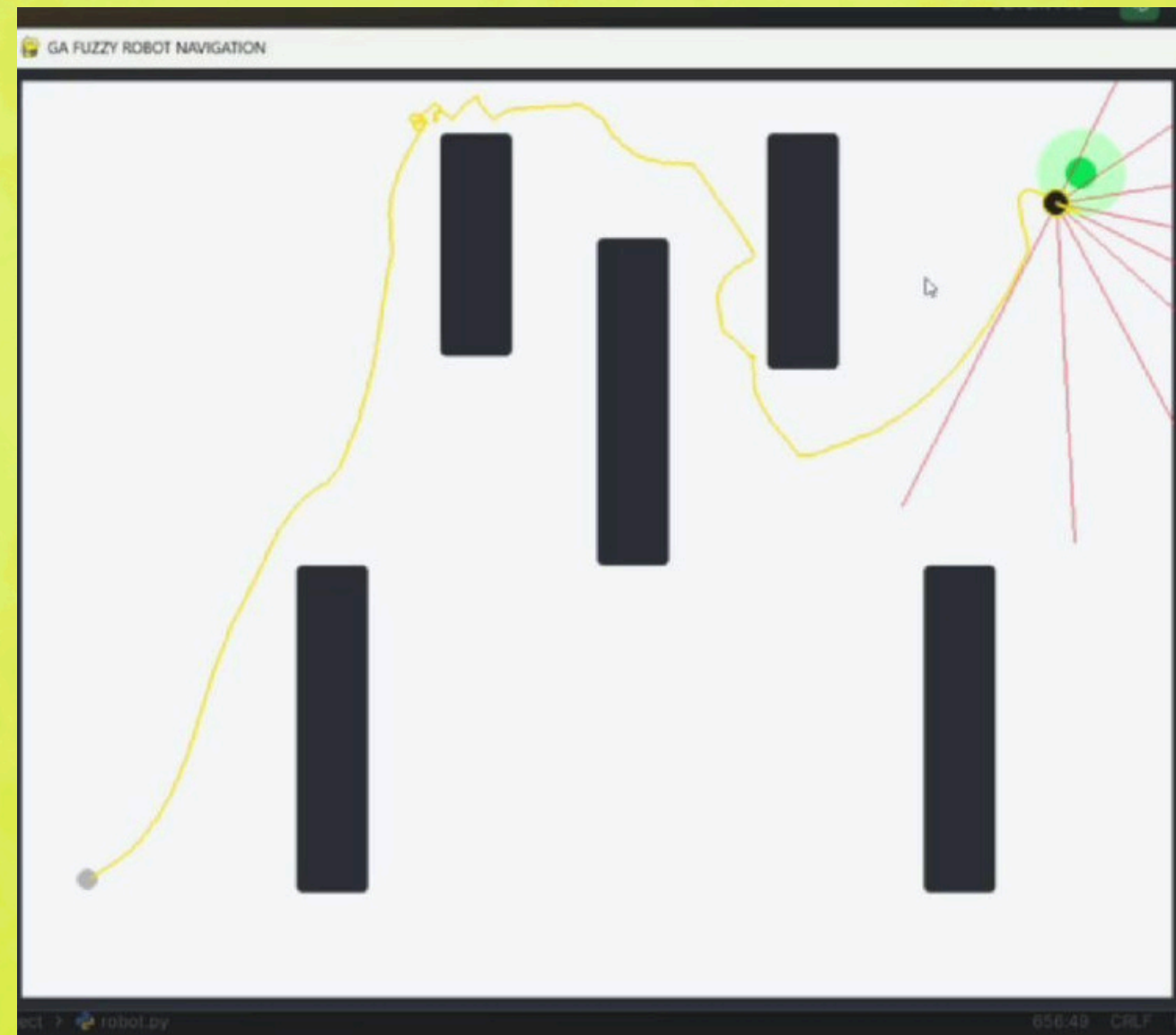


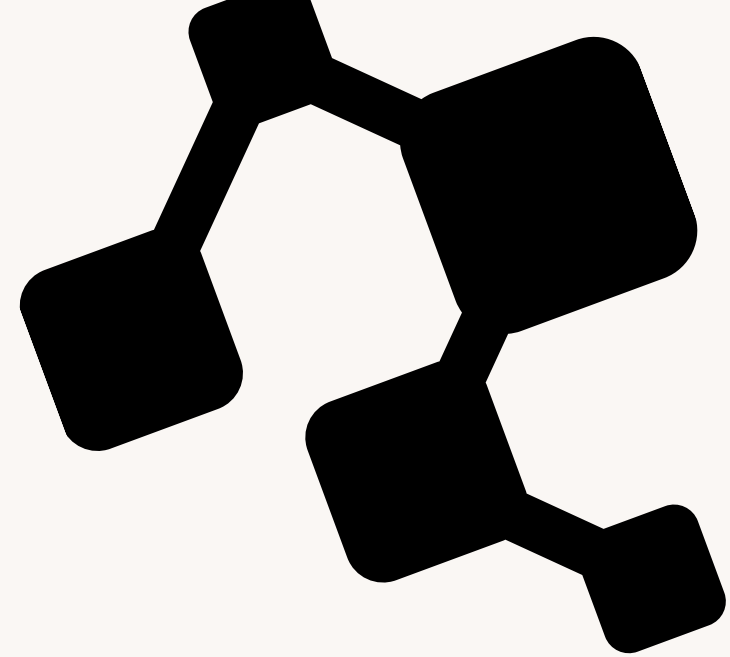
MAP 2



FUZZY LOGIC IN OUR SIMULATION

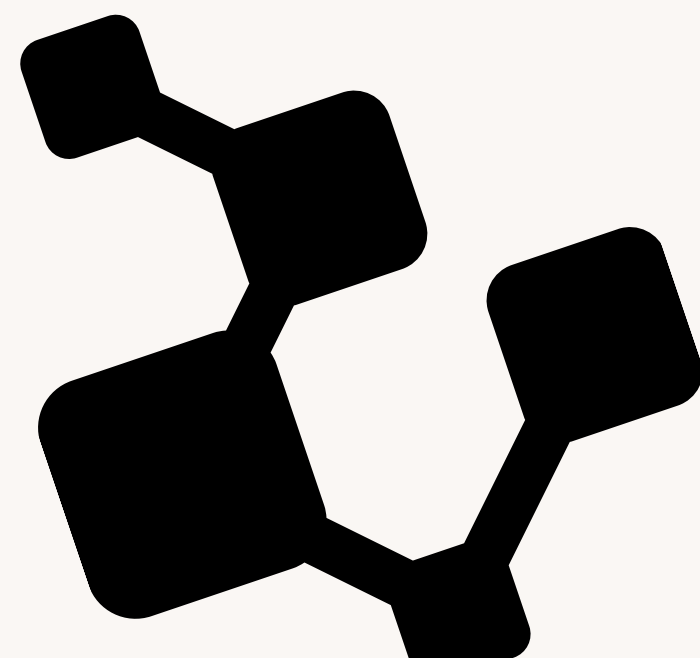
MAP 3





GENETIC ALGORITHM

OPTIMIZATION



THE EVOLUTION PROCESS

1 Population Initialization

Starts with 10 random robot agents.

2 Fitness Evaluation

Judges based on the performance.

3 Selection & Crossover

Best robots DNA strings are mated.

4 Mutation

28% chance to inject new skills.

5 Successful Convergence

Optimized driving parameters send directly to the fuzzy controller.



• CHROMOSOME

1. Sensitivity : Wall detection threshold
2. Turn Power : Steering sharpness intensity
3. Goal Force : Target pulling force
4. Speed : Base velocity of the robot
5. Stuck Escape : Breakaway turning force

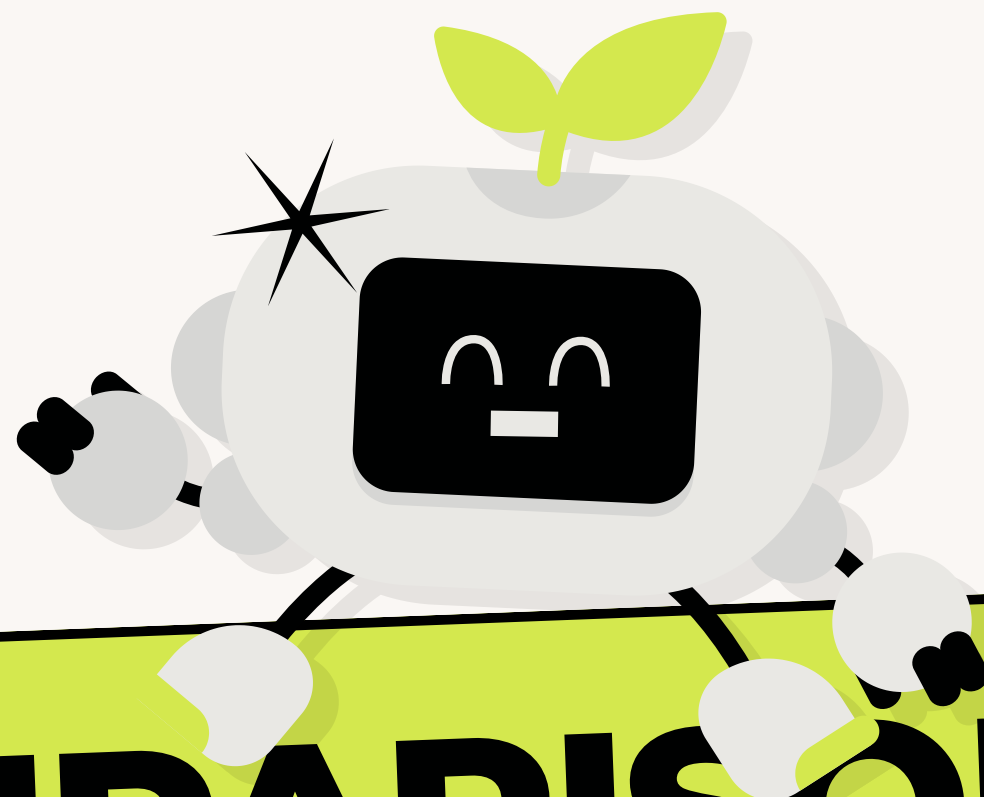
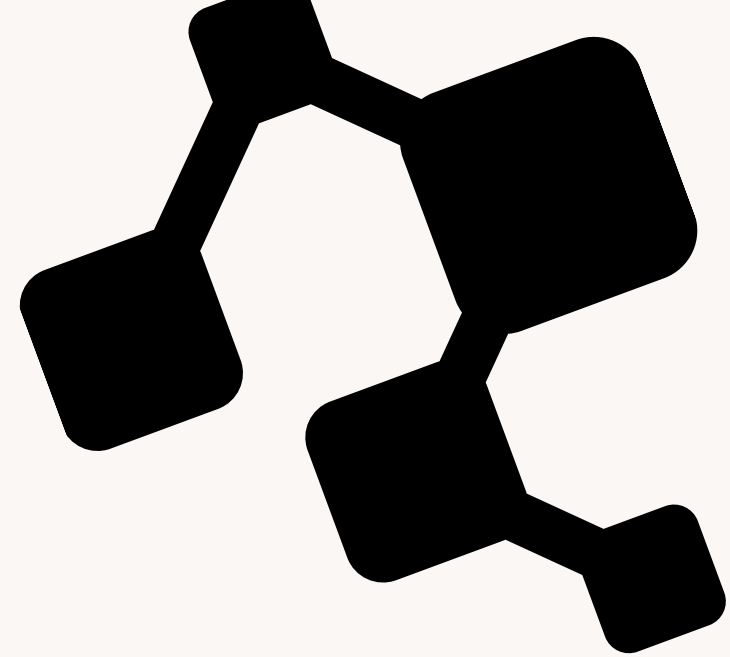
• FITNESS FUNCTION

Robots are mathematically evaluated to reward speed while penalizing every mistake.

$\text{Fitness} = f(\text{Goal time}) - \text{Penalties}$

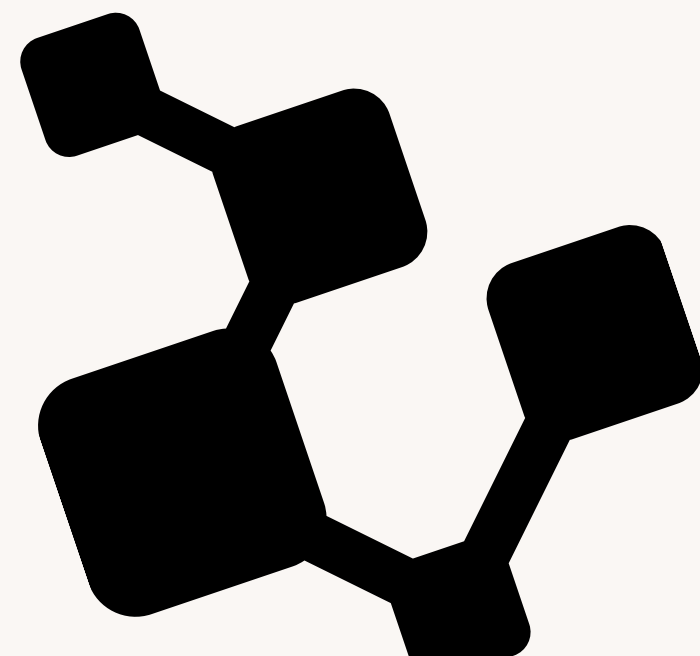
Reward : Reach the goal point quickly.

Penalty : Hitting a wall structure or get stuck.



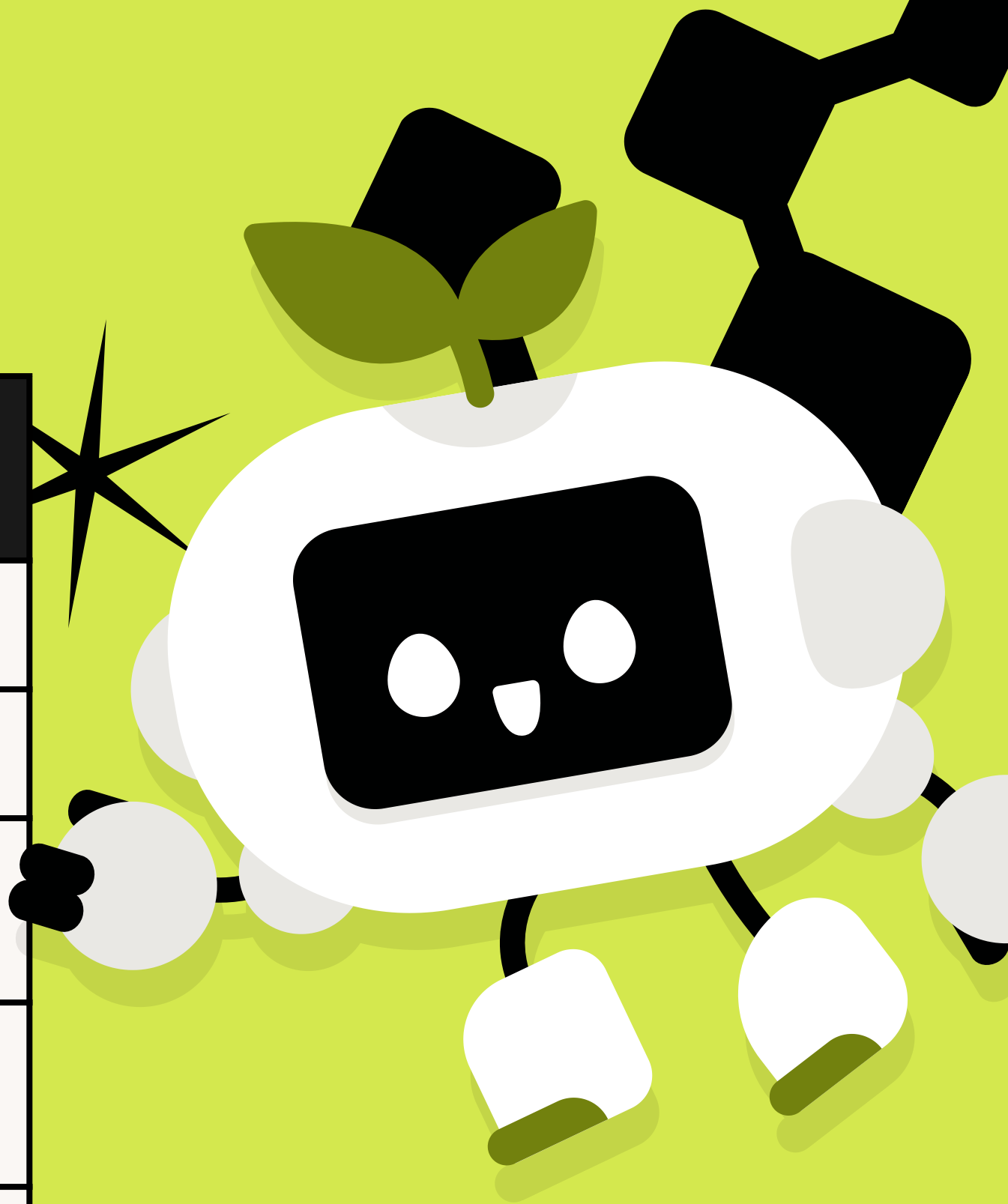
COMPARISON

FUZZY VS FUZZY-GA



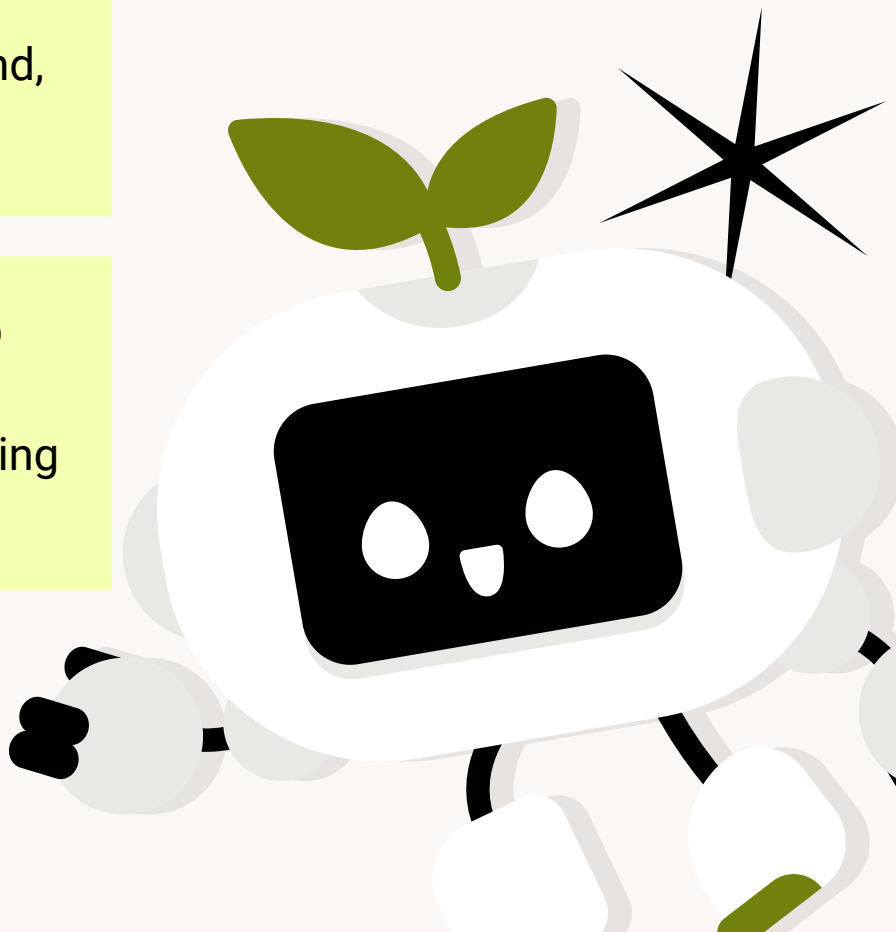
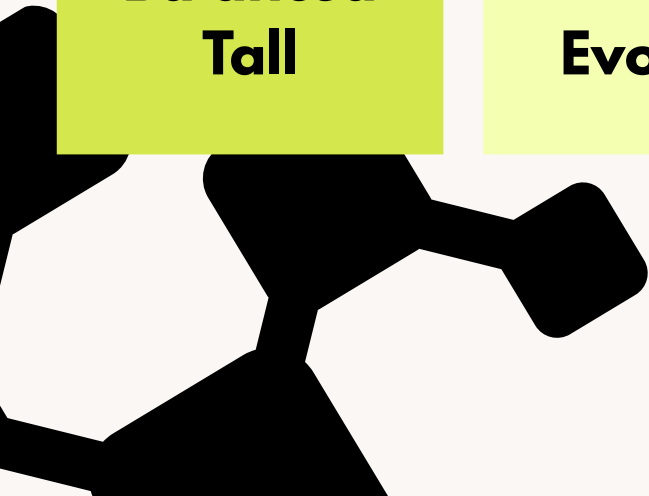
DEFAULT VS EVOLVE

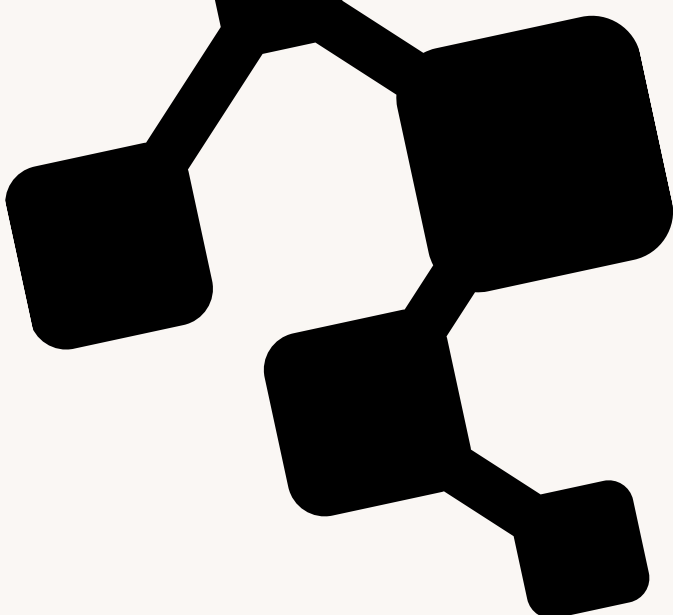
Parameter	Default Mode	Evolve Mode	Comparison	Justification
Sensitivity	138.0	132.4	Decreased	Bolder and smooth path.
Turn Power	0.310	0.222	Decreased	Eliminated over steering.
Goal Force	0.820	0.733	Decreased	Balance navigation to ensure optimal path.
Speed	3.35	3.65	Increased	Higher efficiency of optimized trajectory.
Escape	1.10	1.08	Slightly Decreased	Fine tuned recovery to perform smooth re-direction



DEFAULT VS EVOLVE

Map	Mode	Steps	Collisions	Distance	Behaviour
Map 1 : Balanced Maze	Default	330	0	28.9	The default robot was too scared of walls and turned too early in a jagged, zig-zag way. The evolved robot was braver, driving faster and taking smooth, beautiful corners.
	Evolve	352	1	29.2	
Map 2 : Medium Obstacles	Default	433	1	27.1	The default robot got lucky by taking a wide path. The evolved robot tried a tighter path and got stuck in a dead end, but its succesfully escape.
	Evolve	585	3	29.8	
Map 3 : Balanced Tall	Default	733	8	29.7	Default was very aggressive at the top wall, smashing into it 8 times. GA cleaned up the path completely, reducing steps by half and collapses to only 1.
	Evolve	385	1	27.2	





CONCLUSION

01

GENETIC ALGORITHM WORKS PERFECTLY

The GA successfully found the best tuning settings for the robot's Fuzzy Logic brain, so we do not have to guess the parameters by hand.

02

SMOOTHER AND FASTER DRIVING

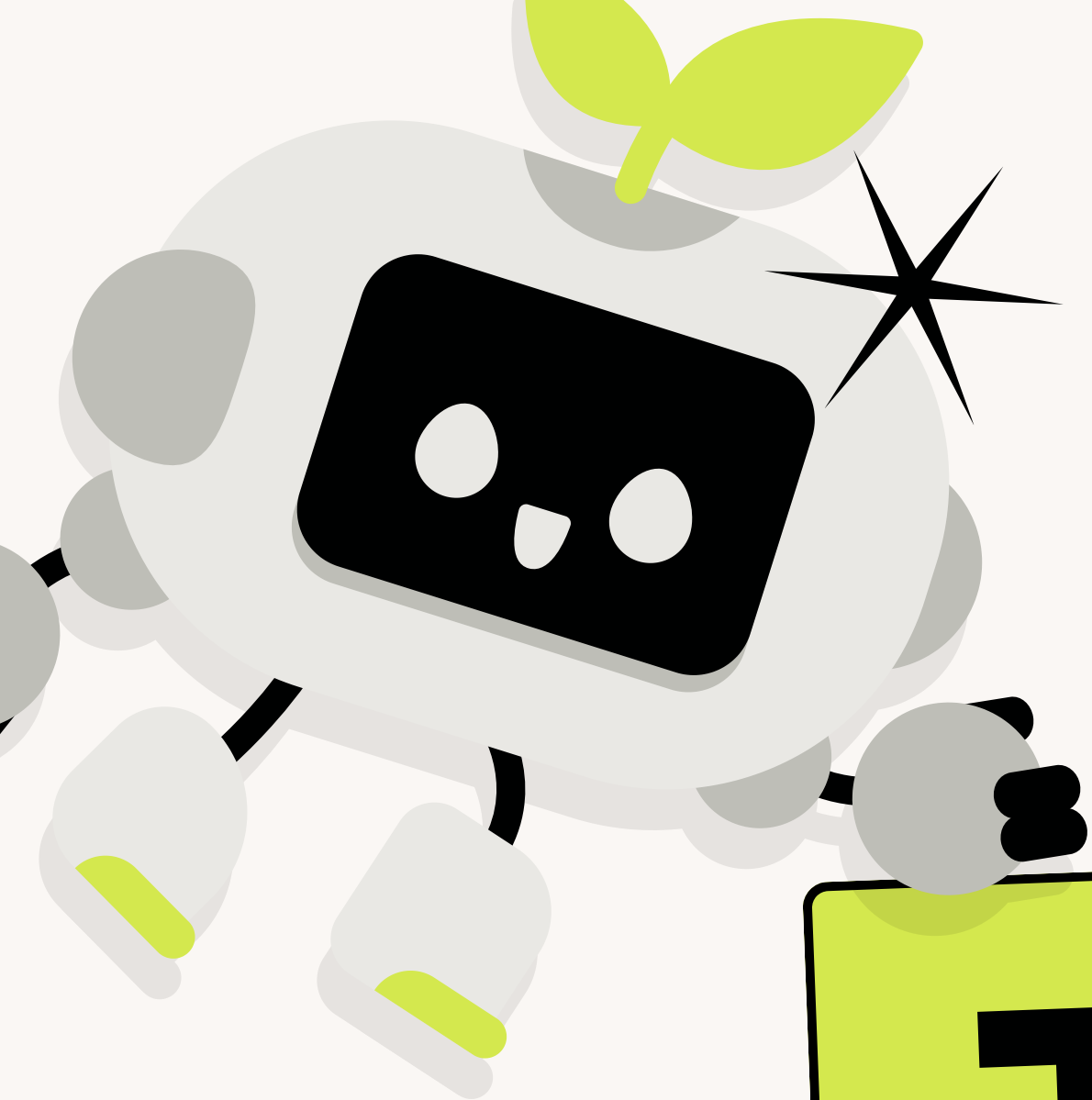
Instead of driving nervously and turning too early, the optimized robot drives smoothly, takes corners beautifully, and moves at a faster speed.

03

SMART AND RELIABLE RECOVERY

Even when the robot makes a mistake or gets trapped in a tough maze, its backup logic successfully helps it untangle itself, escape, and reach the goal safely.





THANK YOU